PGCP-AI

PRACTICAL MACHINE LEARNING

Day 5: Logistic Regression and KNN

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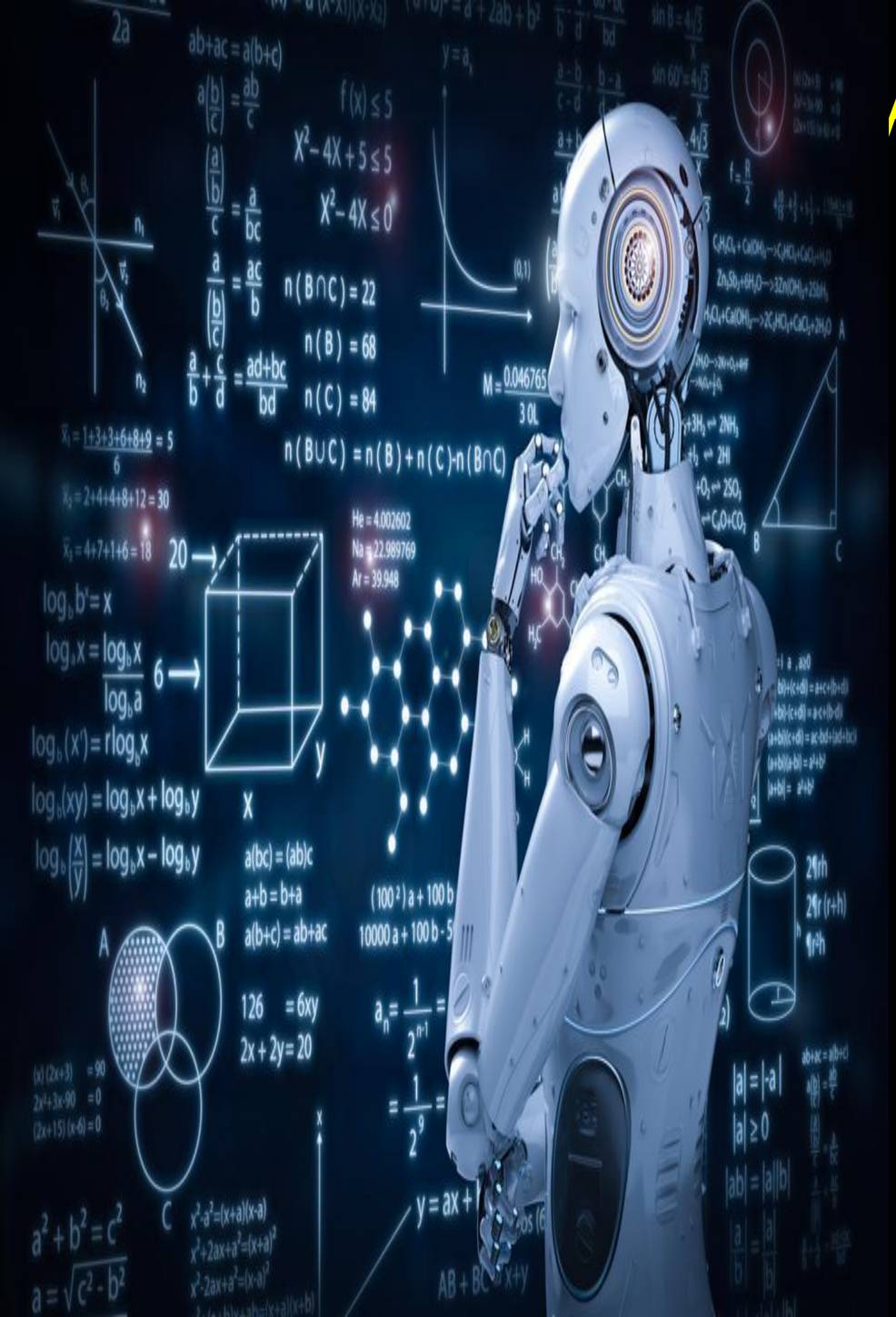
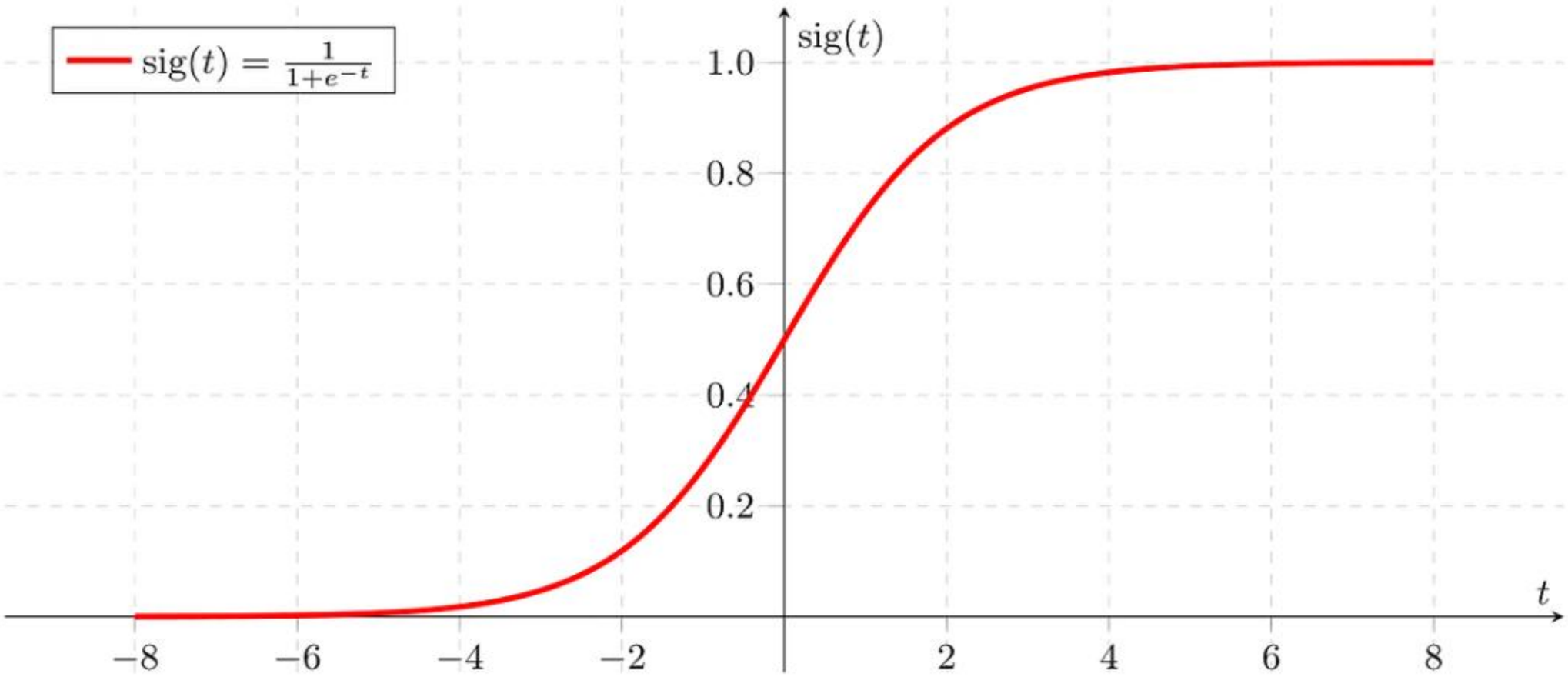


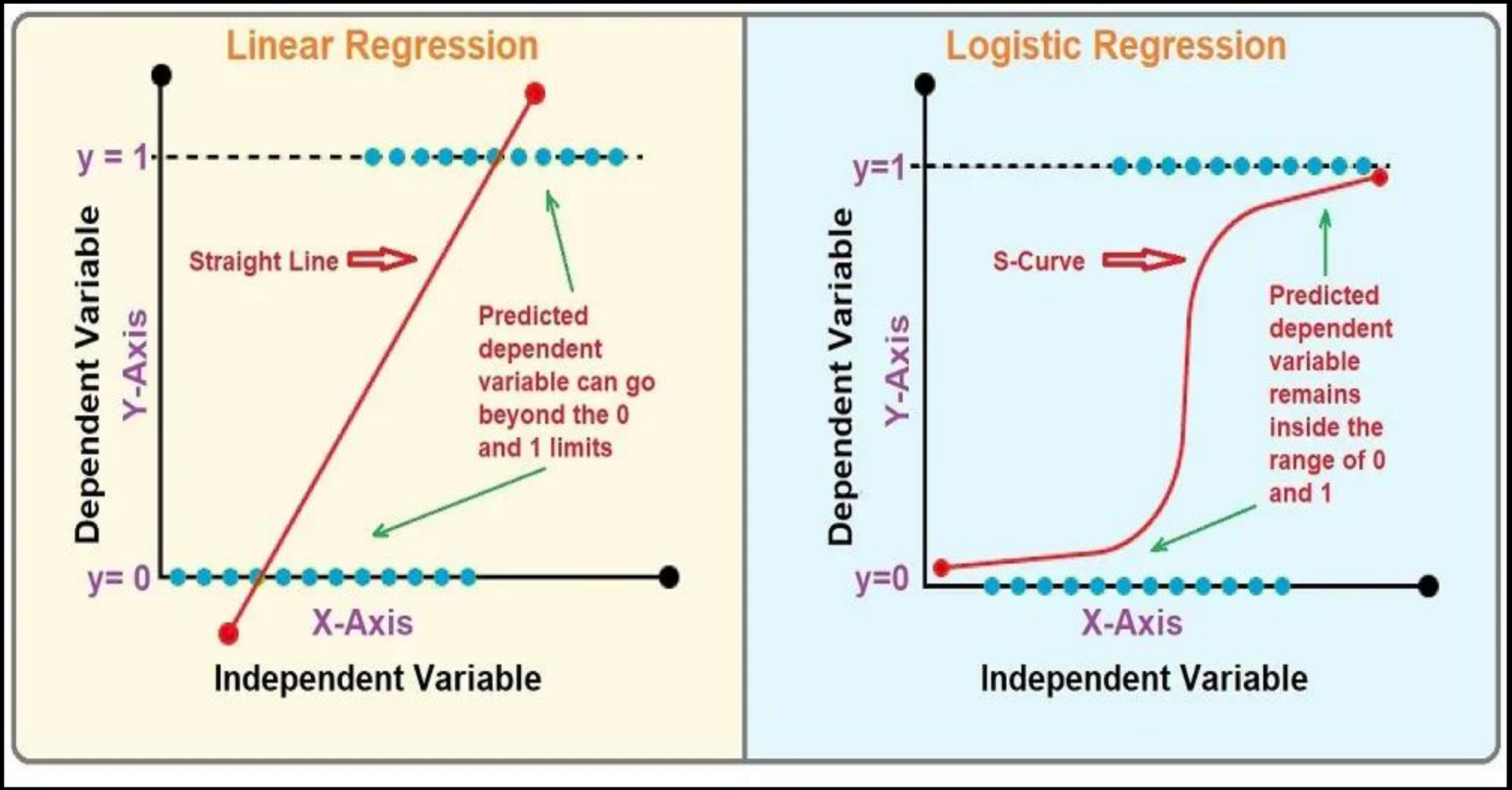
Agenda

- ML Algorithms
- Logistic Regression
- Distance measures
- KNN

Sigmoid Function



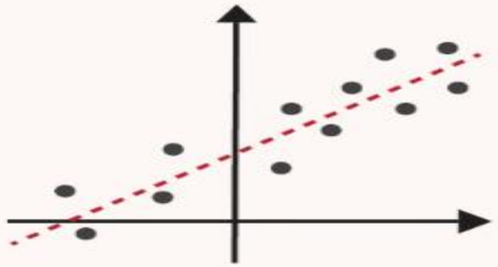
Sigmoid Activation Function


Linear Regression
y = 1
y= 0
Dependent Variable
Y-Axis
Straight Line
Predicted dependent variable can go beyond the 0 and 1 limits
X-Axis
Independent Variable
Logistic Regression
y=1
y=0
Dependent Variable
Y-Axis
S-Curve
Predicted dependent variable remains inside the range of 0 and 1
X-Axis
Independent Variable

Linear regression VS Logistic regression

Linear regression

- Econometric modeling
- Marketing mix model
- Customer lifetime value



Continuous > Continuous

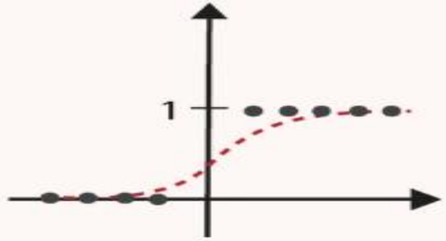
$$y = a_0 + \sum_{i=1}^{N} a_i x_i$$

lm(y-x1 + x2, data)

1 unit increase in
x increases y by a

Logistic regression

- Customer choice model
- Click-through rate
- Conversion rate
- Credit scoring



Continuous > True/False

$$y = \frac{1}{1 + e^{-z}}$$

$$z = a_0 + \sum_{i=1}^{N} a_i x_i$$

glm(y-x1 + x2, data), family = binomial())

1 unit increase in x
increases log odds by a

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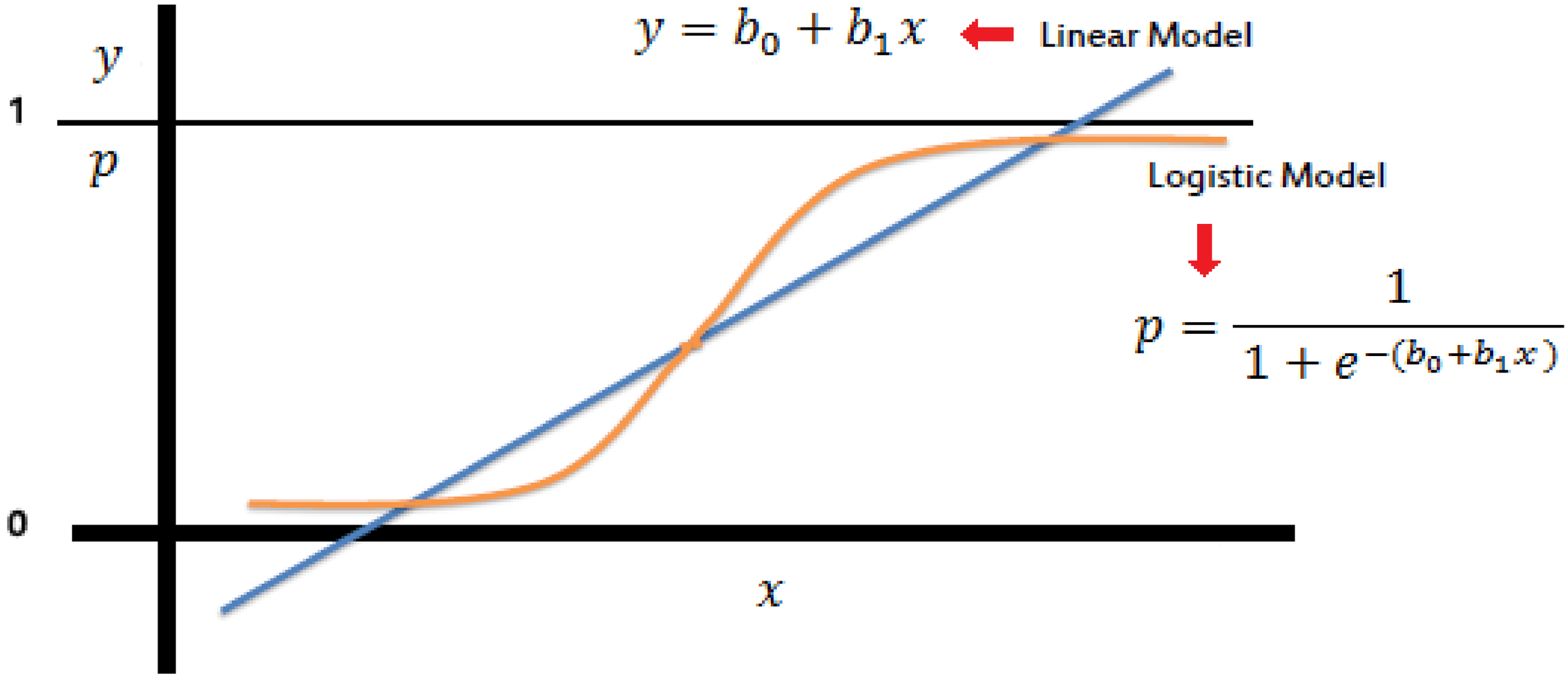

$y = b_0 + b_1 x$
Linear Model
y
1
p
Logistic Model
$p = \frac{1}{1 + e^{-(b_0 + b_1 x)}}$
0
x

Types of Logistic Regression

Binomial

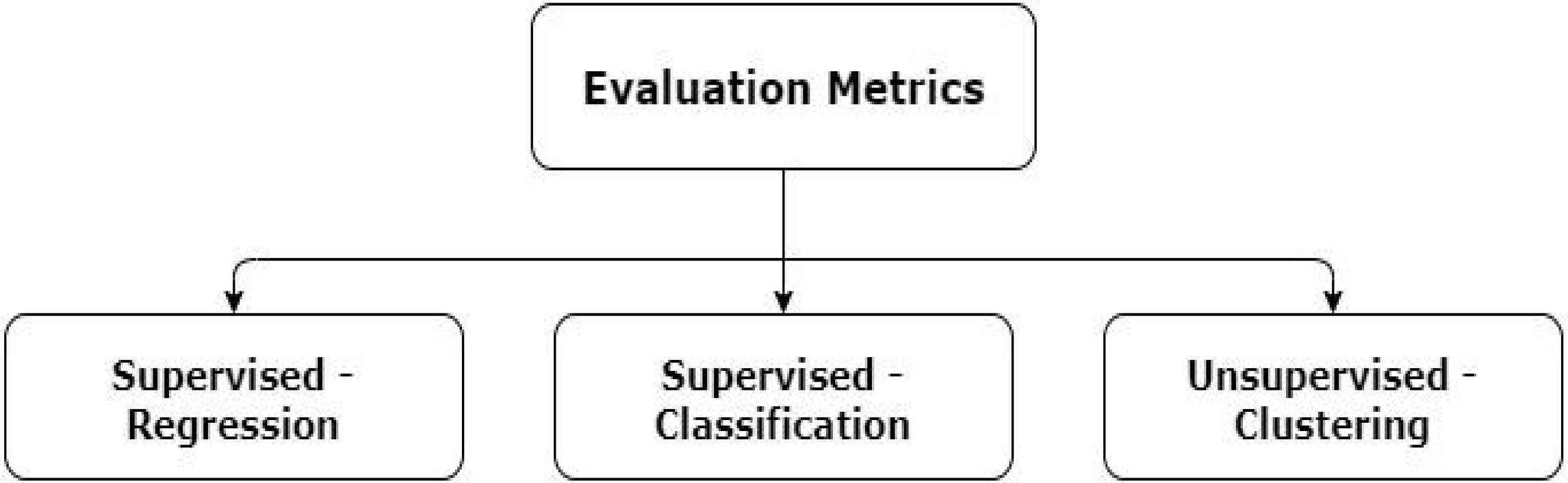
Two classes (0 or 1)

Multinomial

More than two unordered classes (cat, dog, sheep)

Ordinal

More than two ordered classes (low, medium, high)



1. Mean Absolute Error(MAE)
2. Mean Squared Error(MSE)
3. Root Mean Squared Error (RMSE)
4. R squared
5. Adjusted R-Squared

1. Accuracy
2. Precision
3. Recall
4. F1 Score
5. Specificity

1. Rand Index
2. Adjusted Rand Index

Metrics Chart

Metric	Formula	What it Measures	When to Use
Accuracy	(TP + TN) / Total	Overall correctness	Balanced datasets
Precision	TP / (TP + FP)	Correct positive predictions	When FP is costly
Recall (Sensitivity)	TP / (TP + FN)	Ability to find all positives	When FN is costly
F1 Score	2 × (Precision × Recall) / (Precision + Recall)	Balance of precision & recall	Imbalanced datasets
Specificity	TN / (TN + FP)	True negative rate	When negatives matter
ROC-AUC	Area under ROC curve	Class separability	Model comparison
Log Loss	-Σ(y log(p))	Prediction uncertainty	Probabilistic models

Key Formulas (Core Metrics)

$$Precision = \frac{TP}{TP+FP}$$

$$Recall = \frac{TP}{TP+FN}$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Scenario	Best Metric
Balanced dataset	Accuracy
Imbalanced dataset	F1 Score
Avoid false positives	Precision
Avoid false negatives	Recall
Compare models	ROC-AUC
Probabilistic output	Log Loss

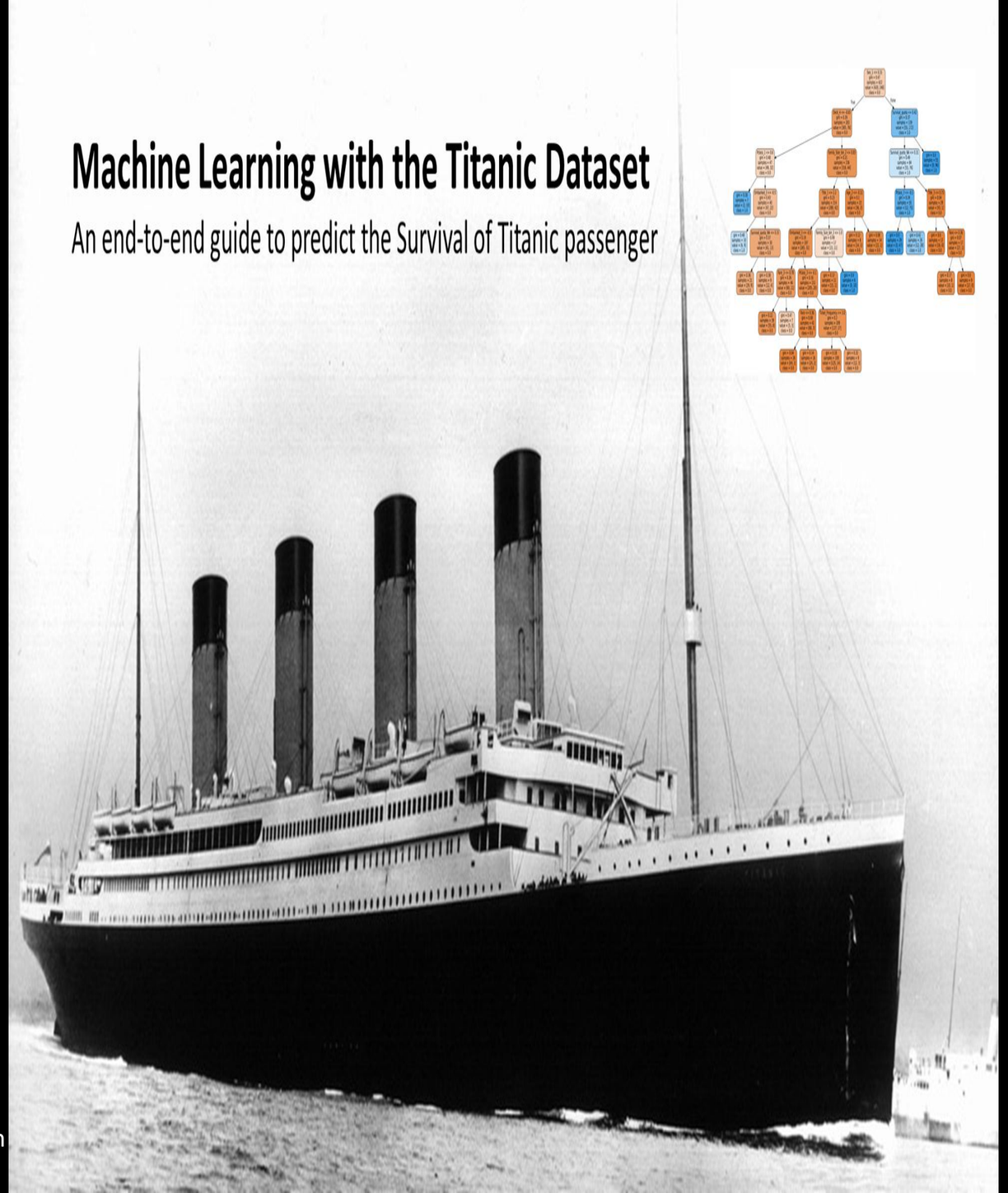
Iris dataset

- Many exploratory data techniques are nicely illustrated with the iris dataset.
  - Dataset created by famous statistician Ronald Fisher
  - 150 samples of three species in genus *Iris* (50 each)
    - *Iris setosa*
    - *Iris versicolor*
    - *Iris virginica*
  - Four attributes
    - sepal width
    - sepal length
    - petal width
    - petal length
  - Species is class label

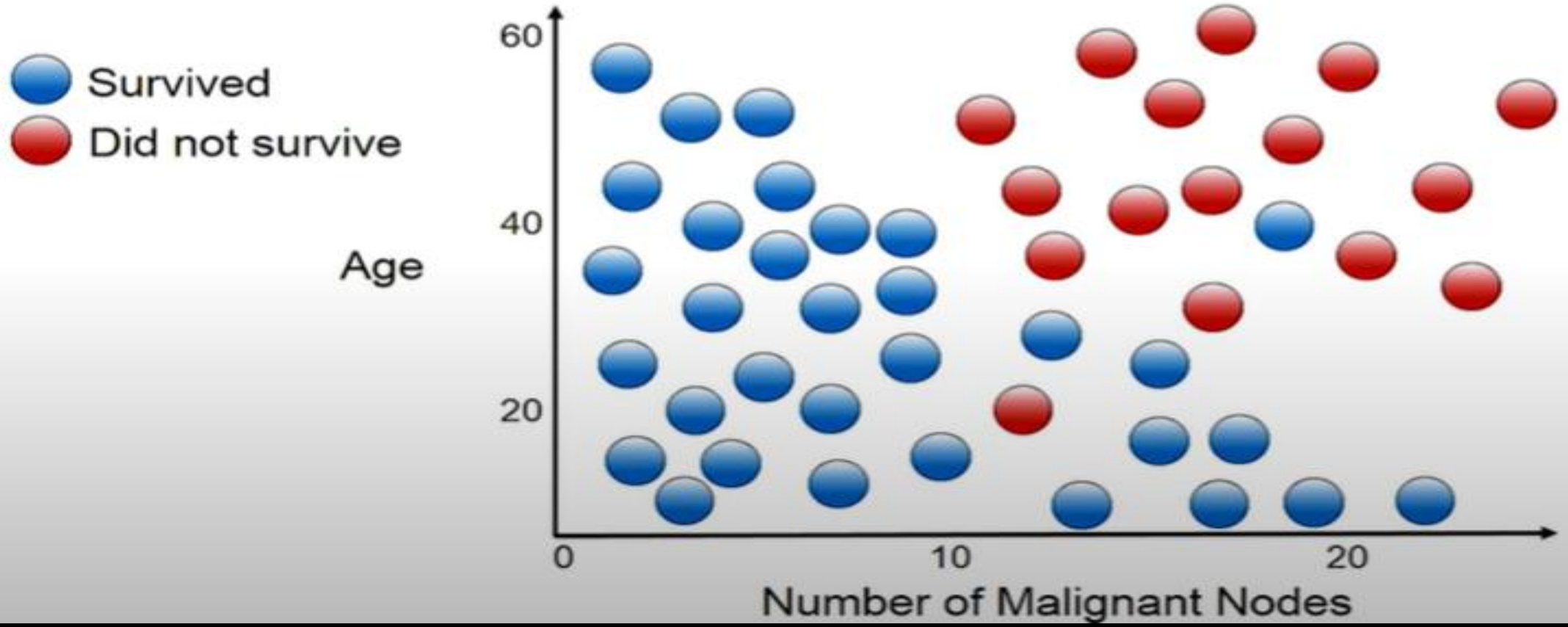
Iris virginica. Robert H. Mohlenbrock. USDA NRCS. 1995. Northeast wetland flora: Field office guide to plant species. Northeast National Technical Center, Chester, PA. Courtesy of USDA NRCS Wetland Science Institute.

Problem Statement

- **Titanic dataset**
- **Explore:** How does each feature relate to whether a person survives/alives?
- Do the EDA in more detail than usual and explain the results!
  - Splitting: 80-20, stratify: y, random_state = 0
- **Preprocessing:**
  - * Drop decks
  - * Fill in the missing value using a simple imputer
  - * One hot encoding: sex, alone
  - * Ordinal encoding: class
  - * Binary encoding: embark town
- **Model selection:**
  - * Evaluation metrics used: F1_score
  - Logistic Regression
  - * Logre, KNN, DecisionTreeClassifier, RandomForestClassifier
  - * Hyperparameter tuning of 2 models that you think is the best
  - * Create a summary for the evaluation results and conclusions on which model is the best for the titanic dataset
- If possible, use pipelines when necessary to avoid data leakage.



K-Nearest Neighbour- Classification



KNN parameters

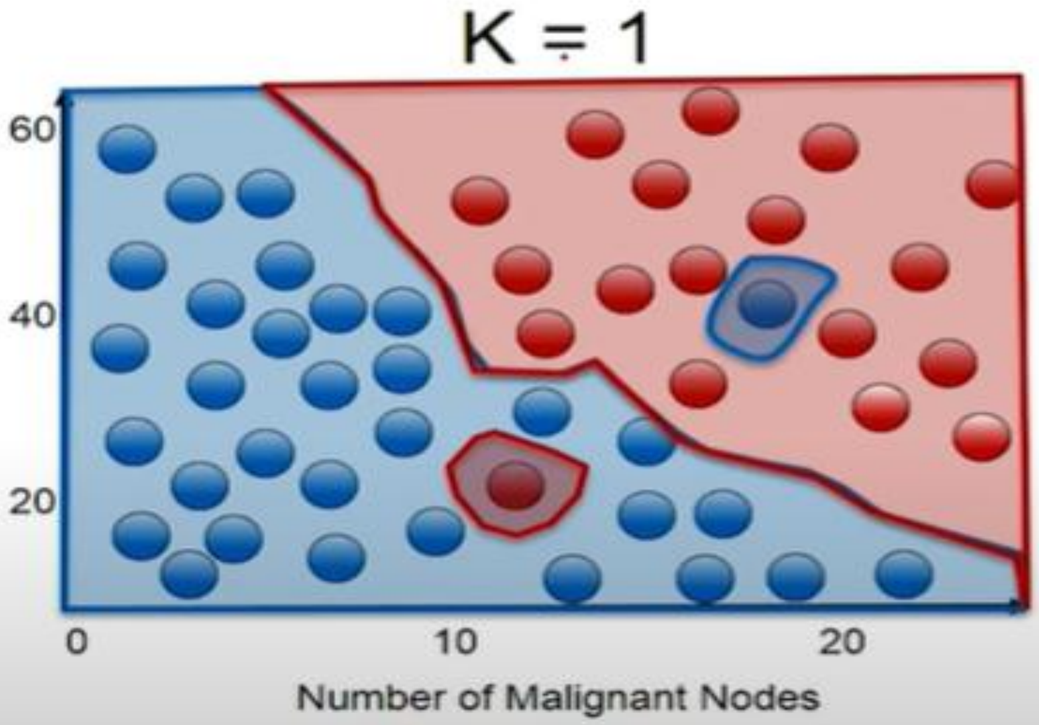
- K – nearest neighbours
- Distance metric

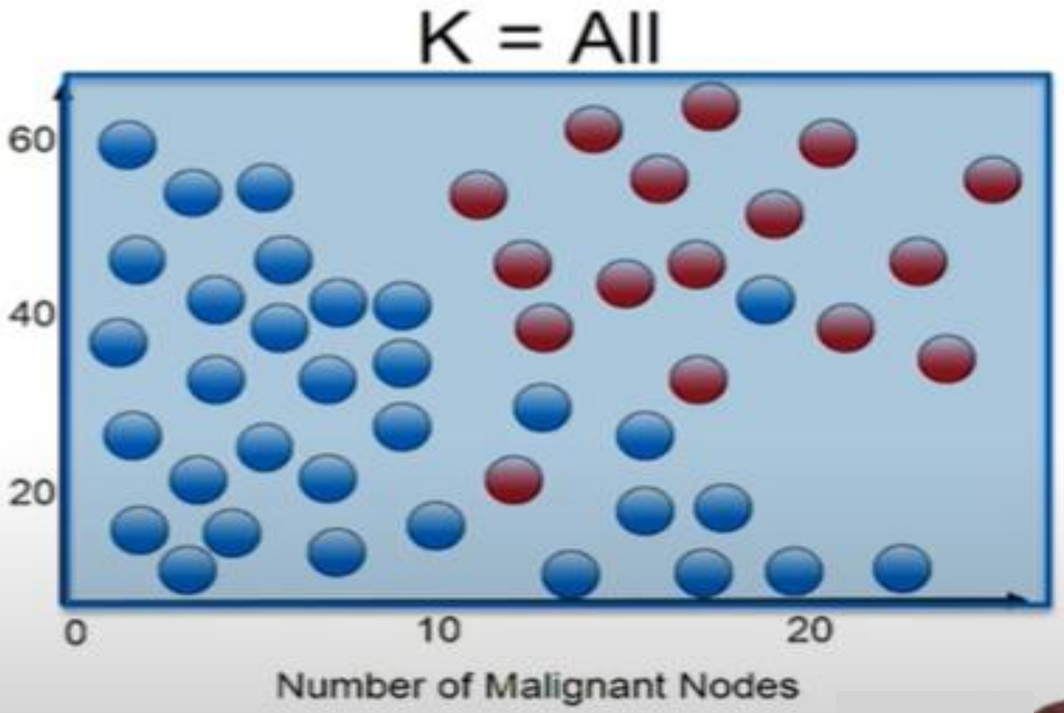
KNN is a **supervised, non-parametric, instance-based (lazy) learning algorithm.**
It does not build an explicit model during training; instead, it stores the dataset and makes predictions by comparing new points to existing ones.

For a new data point:

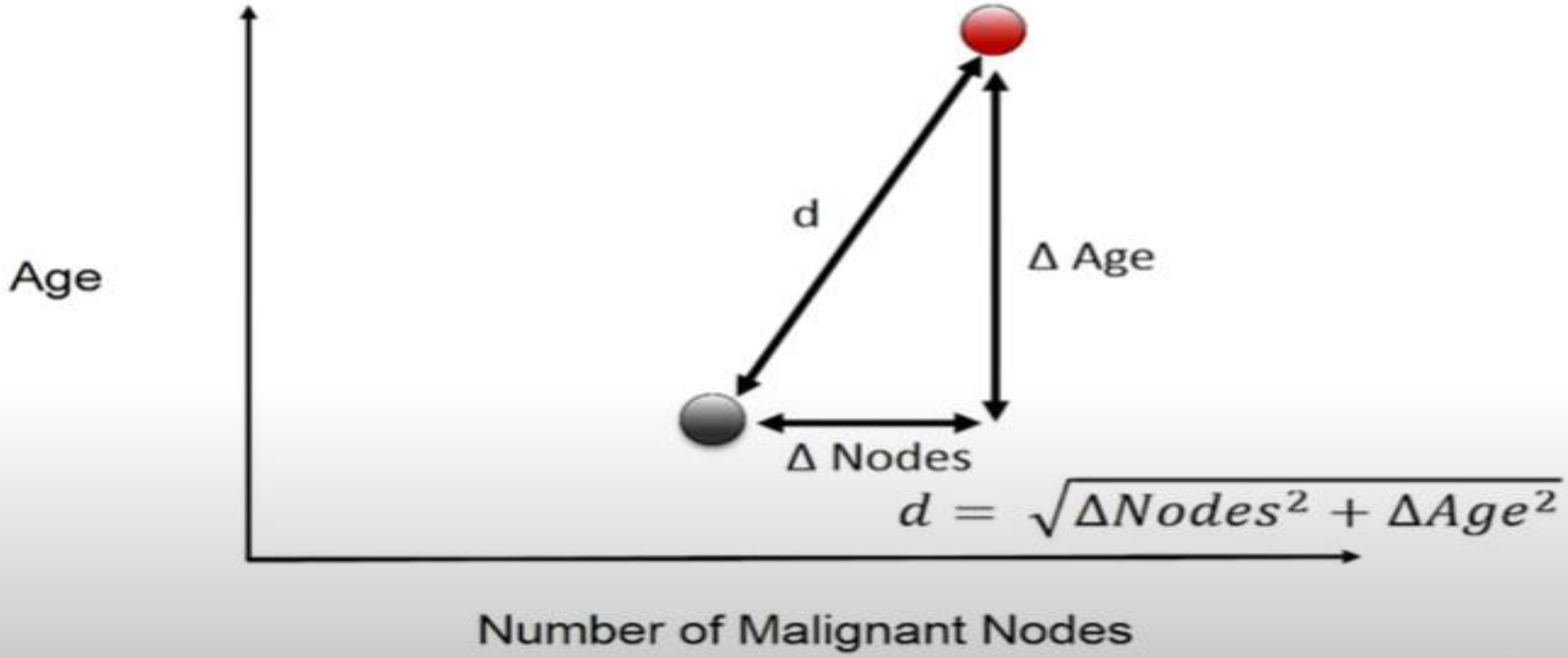
- Find the **K closest points** in the training data
- Use their labels to decide the output

Choosing K


K = 1
60
40
20
0
10
20
Number of Malignant Nodes


K = All
60
40
20
0
10
20
Number of Malignant Nodes

Distance Metric- Euclidean Distance



$$d = \sqrt{\Delta Nodes^2 + \Delta Age^2}$$

Instance based classifiers

Set of Stored Cases

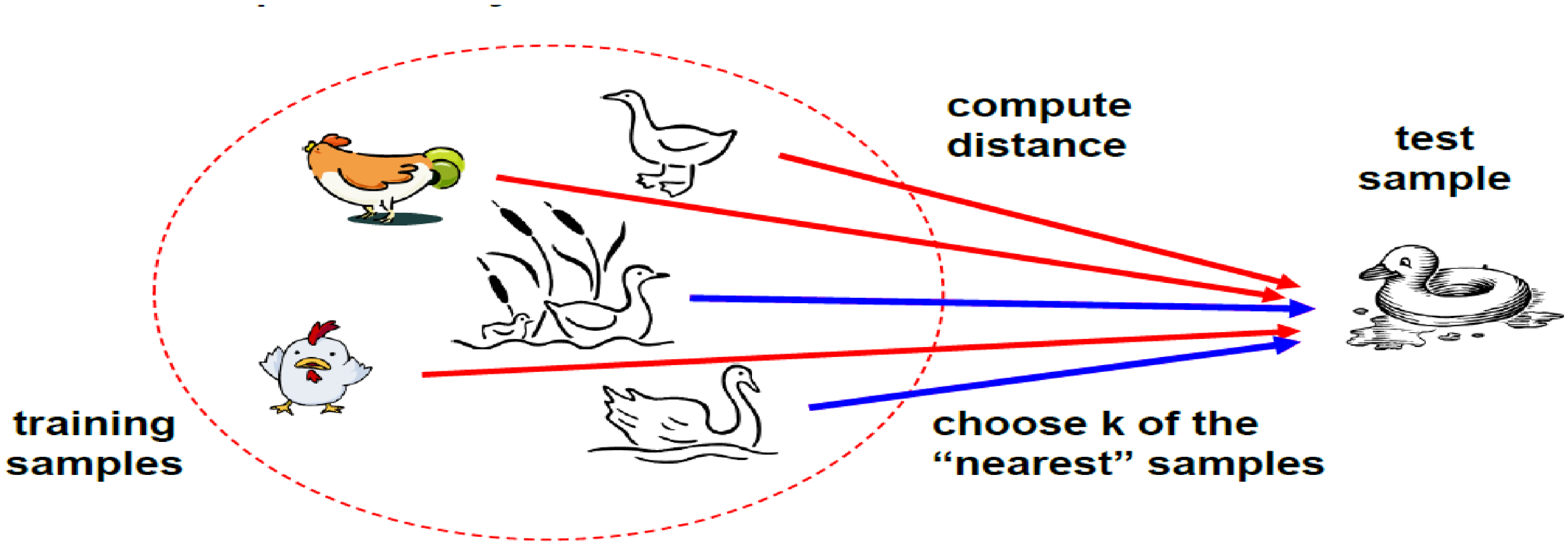
Atr1	……..	AtrN	Class
			A
			B
			B
			C
			A
			C
			B

- **Store the training samples**
- **Use training samples to predict the class label of test samples**

Unseen Case

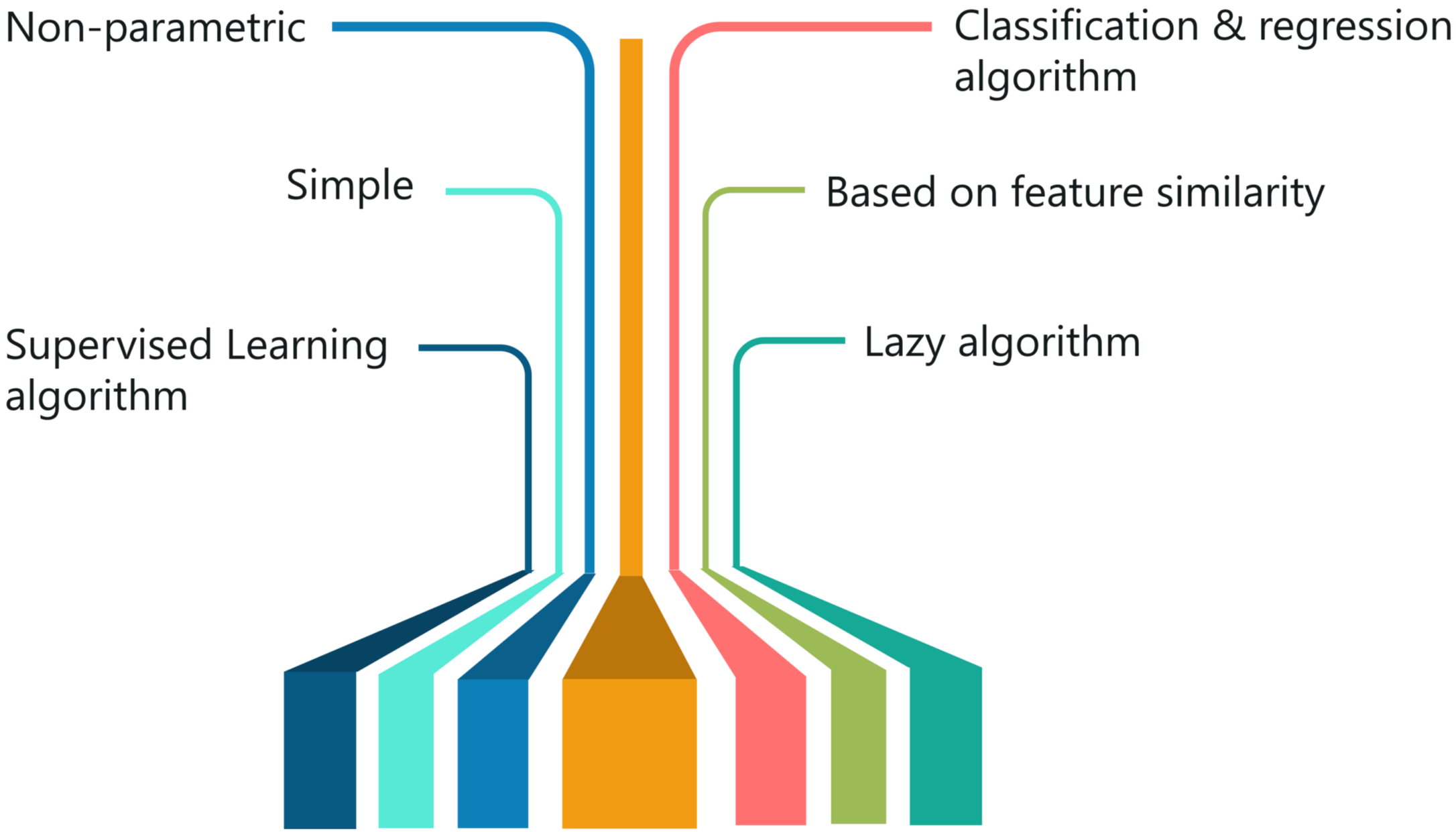
Atr1	……..	AtrN

Nearest Neighbor Classifiers



What is KNN?

- A powerful classification algorithm used in pattern recognition.
- K nearest neighbors stores all available cases and classifies new cases based on a *similarity measure*(e.g **distance function**)
- One of the top data mining algorithms used today.
- A non-parametric lazy learning algorithm (An Instance-based Learning method).


Non-parametric
Classification & regression algorithm
Simple
Based on feature similarity
Supervised Learning algorithm
Lazy algorithm

Lazy Learner

A **lazy learner** is a type of machine learning algorithm that **does not build a model during the training phase.** Instead, it **stores the training data** and performs computation **only when a prediction is required.**

1. Key Idea

- Training phase → **No generalization**
- Prediction phase → **Uses entire dataset**

In simple terms:

> It "learns" only when you ask it to predict.

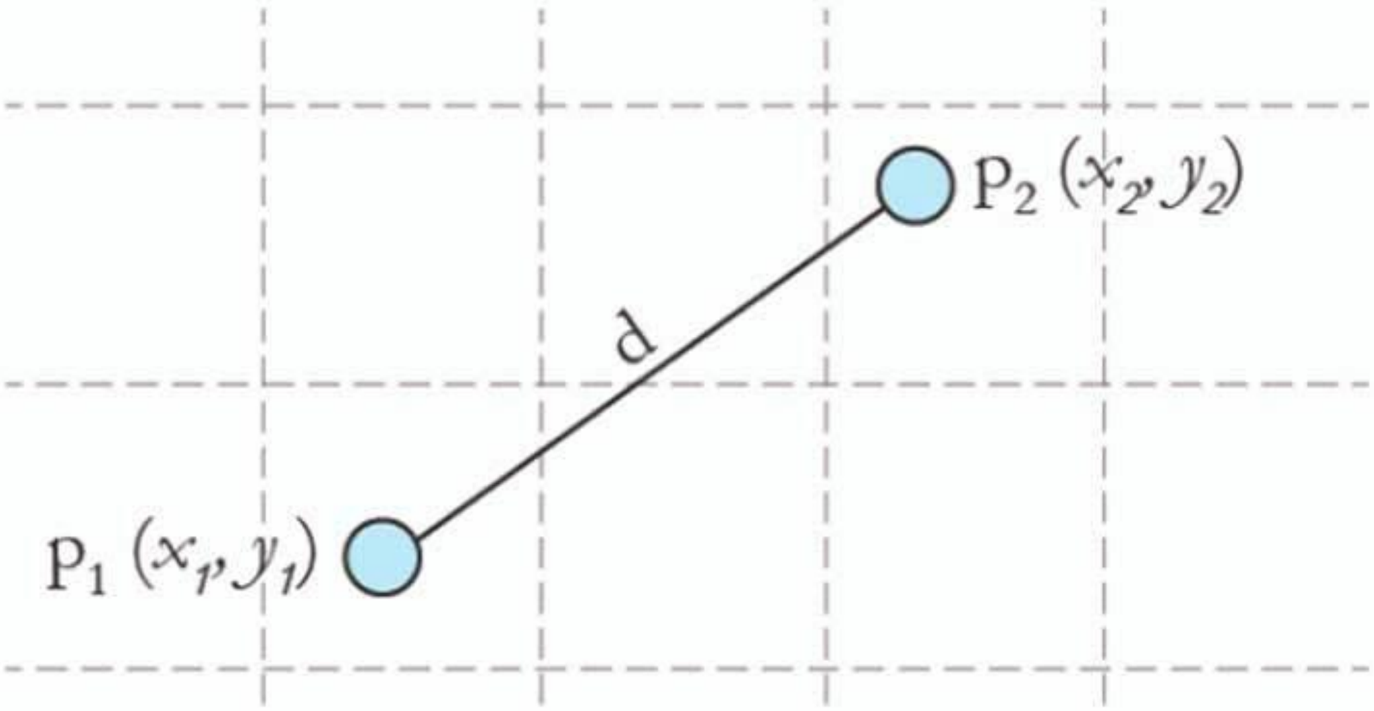
2. How It Works

1. Store all training data
2. When a new input comes:
   - Compare it with stored data
   - Find similar instances
   - Make prediction based on similarity

3. Example Algorithms

- K-Nearest Neighbors (KNN) → most common
- Case-Based Reasoning systems

What is **Euclidean distance?**



$$\text{Euclidean distance (d)} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distance measure for Continuous Variables

Distance functions

Euclidean $$\sqrt{\sum_{i=1}^{k}(x_i - y_i)^2}$$

Manhattan $$\sum_{i=1}^{k}|x_i - y_i|$$

Minkowski $$\left(\sum_{i=1}^{k}(|x_i - y_i|)^q\right)^{1/q}$$

6. Advantages

- Simple to understand and implement
- No training phase
- Works well with small datasets
- Can handle multi-class classification

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7. Disadvantages

- Slow for large datasets
- Sensitive to:
  - Noise
  - Irrelevant features
- Requires **feature scaling**
- Performance drops in high dimensions (**curse of dimensionality**)

8. Importance of Feature Scaling

KNN relies on distance, so features must be scaled:

- Use:
  - Standardization (StandardScaler)
  - Normalization (MinMaxScaler)

Without scaling:

- Larger-value features dominate distance calculation

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9. Simple Example

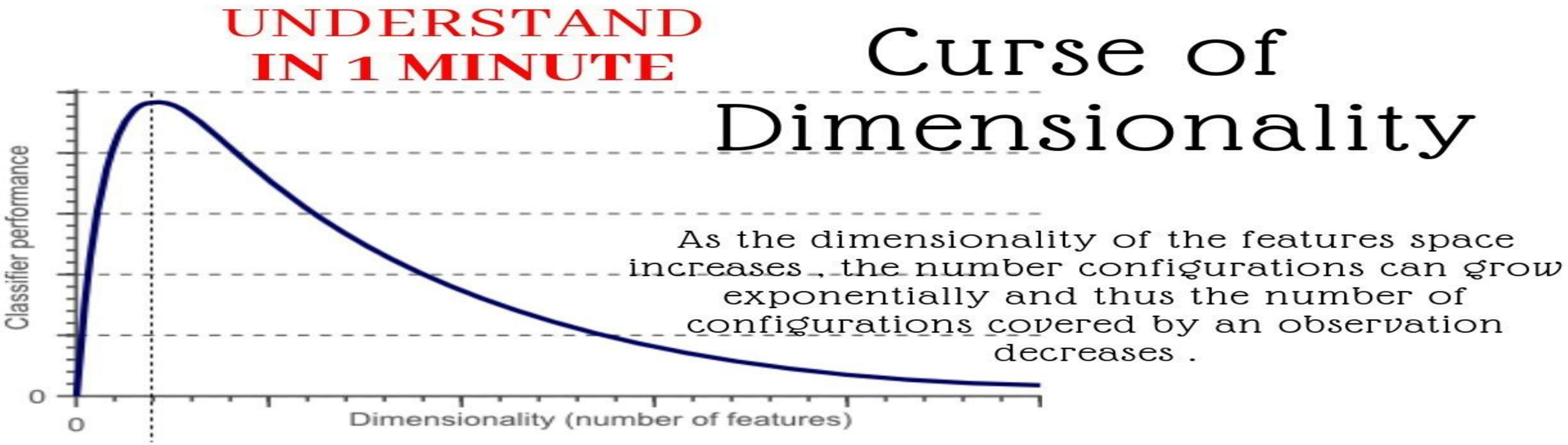
Suppose:

- K = 3
- New point → classify as "A" or "B"

Nearest neighbors:

- A, A, B → majority = **A**

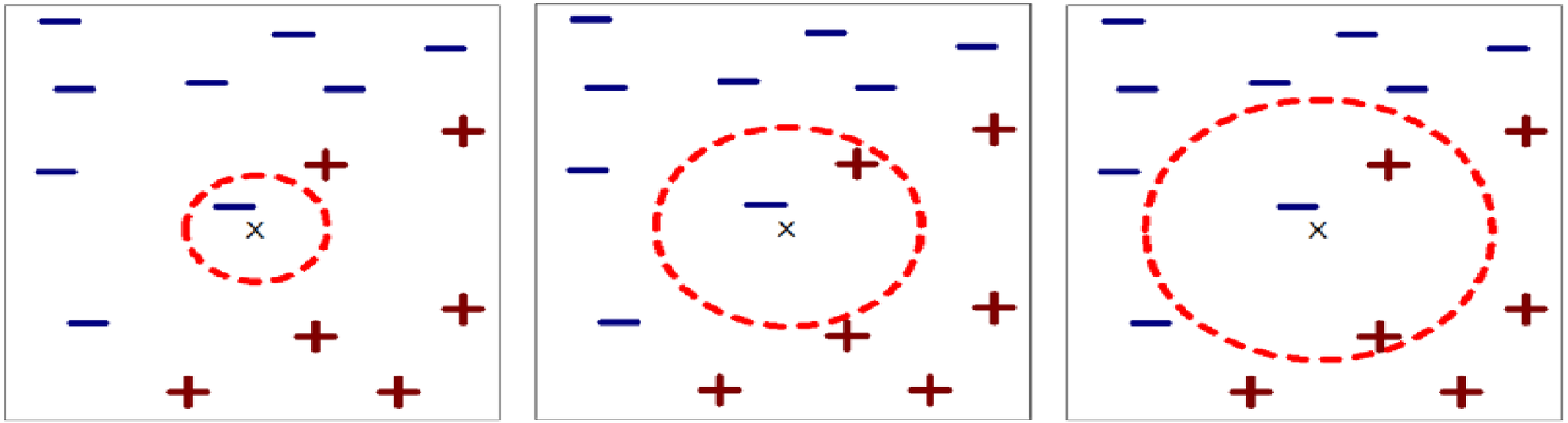
Prediction = **A**



Curse of Dimensionality — Concept

The curse of dimensionality refers to the set of problems that arise when working with high-dimensional data (many features). As the number of dimensions increases, data becomes sparse, and many machine learning algorithms become less effective.

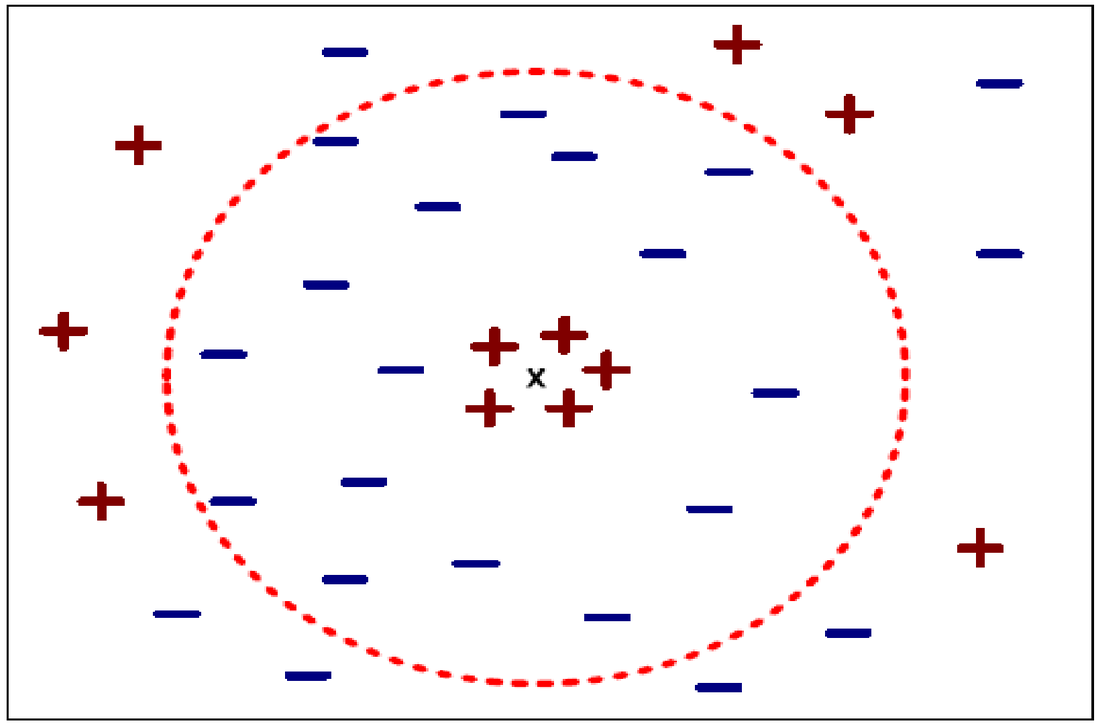
Predicting class from nearest neighbors



nearest neighbors	1	2	3
majority vote	–	?	+
distance-weighted vote	–	–	– or +

Predicting class from nearest neighbors

- Choosing the value of *k*:
  - If *k* is too small, sensitive to noise points
  - If *k* is too large, neighborhood may include points from other classes



K-Nearest Neighbor Algorithm

- All the instances correspond to points in an n-dimensional feature space.
- Each instance is represented with a set of numerical attributes.
- Each of the training data consists of a set of vectors and a class label associated with each vector.
- Classification is done by comparing feature vectors of different K nearest points.
- Select the K-nearest examples to E in the training set.
- Assign E to the most common class among its K-nearest neighbors.

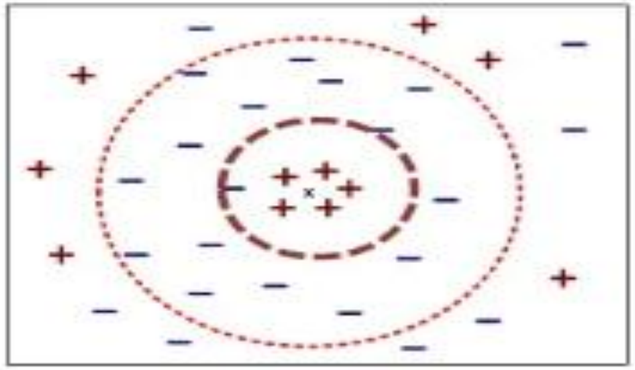
How does K-NN work?

The K-NN working can be explained on the basis of the below algorithm:

- **Step-1:** Select the number K of the neighbors
- **Step-2:** Calculate the Euclidean distance of **K number of neighbors**
- **Step-3:** Take the K nearest neighbors as per the calculated Euclidean distance.
- **Step-4:** Among these k neighbors, count the number of the data points in each category.
- **Step-5:** Assign the new data points to that category for which the number of the neighbor is maximum.
- **Step-6:** Our model is ready.

How to choose K?

- If K is too small it is sensitive to noise points.
- Larger K works well. But too large K may include majority points from other classes.



- Rule of thumb is K < sqrt(n), n is number of examples.

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KNN Feature Weighting

- Scale each feature by its importance for classification

$$D(a,b) = \sqrt{\sum_k w_k (a_k - b_k)^2}$$

- Can use our prior knowledge about which features are more important
- Can learn the weights $\mathbf{w_k}$ using **cross-validation** (to be covered later)

Feature Normalization

- Distance between neighbors could be dominated by some attributes with relatively large numbers.
  - e.g., income of customers in our previous example.

$$a_i = \frac{v_i - \min v_i}{\max v_i - \min v_i}$$

- Arises when two features are in different scales.

- Important to normalize those features.
  - Mapping values to numbers between $0 - 1$.

Strengths of KNN

- Very simple and intuitive.
- Can be applied to the data from any distribution.
- Good classification if the number of samples is large enough.

Weaknesses of KNN

- Takes more time to classify a new example.
  - need to calculate and compare distance from new example to all other examples.
- Choosing k may be tricky.
- Need large number of samples for accuracy.

KNN Classification – Distance

Age	Loan	Default	Distance
25	$40,000	N	102000
35	$60,000	N	82000
45	$80,000	N	62000
20	$20,000	N	122000
35	$120,000	N	22000
52	$18,000	N	124000
23	$95,000	Y	47000
40	$62,000	Y	80000
60	$100,000	Y	42000
48	$220,000	Y	78000
33	$150,000	Y	8000
48	**$142,000**	**?**	

Euclidean Distance

$$D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Problem Statement

Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

Identify the Play Golf possibility if Outlook=‘Overcast’, Temp=‘Cool’, Wicndy=‘True’.

THANK YOU